

Brhane Teamrat Gidey

LLM and Agentic AI Researcher | AI for Software Engineering

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Research Profile

LLM and agentic AI researcher with more than six years of software engineering experience and an MSc specialization in reliable code generation. Conducting thesis research at York University on whether structured, verifiable behavioural specifications can improve the functional correctness of smaller code language models and enable more targeted refinement than unstructured prompting or execution feedback alone. Experienced in LoRA fine-tuning, automated testing, failure analysis, retrieval-augmented generation, and reproducible model evaluation. Combines research experience in reliable LLM reasoning and agentic refinement with extensive production experience in backend, cloud, healthcare, and AI systems.

Research Interests

Large language models; agentic AI; reliable and trustworthy model reasoning; code language models; structured and verifiable generation; LLM fine-tuning and adaptation; execution-based evaluation; retrieval-augmented generation; multimodal learning; AI for software engineering.

Education

MSc in Artificial Intelligence and Computer Science

October 2024 – Expected October 2026

University of Calabria

Italy

Thesis: *Contract-Guided LLM Code Generation*

Research placement: Thesis research conducted at York University, Canada, through the Erasmus+ MoST International Mobility Scholarship.

Selected coursework: Intelligent Agents, Deep Learning, Optimization for Machine Learning, Statistical Methods for Data Science, Research and Development Methodologies, Theoretical Computer Science, and GPU Programming.

BSc in Computer Science and Engineering

October 2014 – July 2019

Mekelle University, Mekelle Institute of Technology

Ethiopia

Selected coursework: Data Structures and Algorithms, Software Engineering, Database Management Systems, Operating Systems, Neural Networks, Artificial Intelligence and Expert Systems, and Object-Oriented Programming in C++.

Research Experience

MSc Thesis Researcher – Contract-Guided LLM Code Generation - Reliable Code Language Models

March 2026 – Present

York University / University of Calabria

Canada / Italy

- Investigating whether structured and verifiable behavioural contracts can improve code-LLM correctness and support more targeted refinement than natural-language prompting, testing, or execution feedback used independently.
- Designed a contract-guided workflow in which natural-language requirements are transformed into preconditions, postconditions, and invariants that support code generation, automated validation, failure diagnosis, repair, and re-verification.
- Constructed a task-to-contract corpus containing behavioural specifications, task metadata, validation outcomes, execution traces, and structured failure information for supervised fine-tuning, preference learning, and controlled evaluation.
- Fine-tuned and evaluated DeepSeek, Qwen, Mistral, and Llama-family models using PyTorch, Hugging Face Transformers, and PEFT/LoRA; compared adapted models with untuned checkpoints, larger same-family models, and frontier baselines.
- Built an agentic refinement workflow that generates solutions, executes tests, classifies failures, selects repair strategies, regenerates or modifies code, and re-verifies corrected outputs.
- Implemented execution-based evaluation pipelines for HumanEval, BigCodeBench, ClassEval, LiveCodeBench, and TACO, including runtime contract checking, no-regression validation, and analysis of logical, runtime, timeout, signature, interface, and incomplete-implementation failures.
- Extended refinement through contextual bandits, reinforcement-learning-based decision policies, and Direct Preference Optimization to study failure-aware repair and model adaptation.

Research Outputs

- **Contract-Guided LLM Code Generation** — MSc thesis, expected 2026.

Selected AI Projects

Medical Knowledge and Healthcare AI Assistant

Python, FastAPI, RAG, BM25, FAISS, Llama

Developed a source-grounded healthcare assistant over WHO and CDC documents using hybrid keyword and dense retrieval, evidence filtering, multi-step reasoning, and citation-supported answer generation. Exposed real-time and streaming queries through a containerized FastAPI service.

Automatic Medical Imaging Report Generator

PyTorch, ResNet-50, GPT-2, Transformers

Developed an end-to-end multimodal system that maps chest X-ray visual embeddings into a language-model representation space for clinical-style report generation; performed data preparation, fine-tuning, validation, and quantitative error analysis.

Professional Experience

Freelance Software and AI Engineer

December 2022 – Present

Upwork and Freelancer.com

Remote / International

- Delivered production healthcare, finance, enterprise, web, mobile, and AI systems for clients across North America, Europe, Asia, and the Middle East.
- Designed and deployed REST APIs, microservices, multi-tenant platforms, database architectures, authentication systems, integrations, AI workflows, and cloud-ready applications.
- Translated ambiguous requirements into maintainable architectures, implementation plans, tested features, documentation, demonstrations, and deployable production systems.
- **Completed more than 3,360 Upwork hours while maintaining Top Rated status and a 100% Job Success Score, along with five-star feedback on Freelancer.com.**

Software Engineer

March 2020 – September 2023

Medco Technology Solutions

Addis Ababa, Ethiopia

- Led development and maintenance of electronic medical-record, laboratory-information, electronic-prescription, pharmacy, inventory, billing, reporting, and administrative systems.
- Designed backend services, REST APIs, relational database schemas, role-based access controls, frontend modules, reports, and third-party integrations from requirements analysis through production deployment.
- Diagnosed production defects, reviewed code changes, supported regression testing, validated databases, mentored junior engineers, and improved delivery through structured defect triage and release support.
- Delivered an integrated EMR and laboratory platform that remains in active clinical use.

ICT Consultant

September 2019 – March 2020

Nayna Saving and Credit Cooperative

Ethiopia

- Advised management on IT strategy, infrastructure, security, backup and recovery, software support, process digitization, and operational reliability while supervising two IT staff members.

Technical Expertise

Large language models and agentic AI: Code LLMs, LLM-based code generation, coding agents, agentic workflows, retrieval-augmented generation, structured reasoning, multi-step refinement, generated-code evaluation, and reliable failure handling.

Model adaptation and learning: PyTorch, TensorFlow, Hugging Face Transformers, PEFT, LoRA, supervised fine-tuning, contextual bandits, reinforcement learning, Direct Preference Optimization, DeepSeek, Qwen, Mistral, and Llama.

Specifications and evaluation: Design by Contract, preconditions, postconditions, invariants, runtime contract checking, automated testing, test generation, program repair, failure taxonomy, regression testing, no-regression validation, benchmark evaluation, and controlled comparative experiments.

Retrieval and research engineering: BM25, FAISS, vector databases, semantic retrieval, evidence filtering, dataset construction, data validation, GPU training, Linux, Git, Docker, reproducible experiment pipelines, and technical documentation.

Programming and software systems: Python, Java, JavaScript, TypeScript, C, C++, PHP, Dart, SQL, Spring Boot, FastAPI, Laravel, React, Vue.js, Flutter, REST APIs, microservices, multi-tenancy, Docker, AWS, CI/CD, PostgreSQL, MySQL, MongoDB, Redis, Neo4j, and Apache Spark.

Selected Production Systems

Led or contributed to production systems including an integrated electronic medical-record and laboratory-information platform, electronic prescription and pharmacy-management systems, multi-tenant restaurant-management software, food-ordering and delivery platforms, mobile applications, and cloud-based enterprise services using Spring Boot, Laravel, Vue.js, Flutter, PostgreSQL, MySQL, Docker, AWS, and REST APIs.

Honors and Scholarships

- Erasmus+ MoST International Mobility Scholarship for MSc thesis research at York University.
- Erasmus+ G.R.E.A.T. Traineeship Scholarship.
- University of Calabria Extra-EU Master's Admission Scholarship.
- Undergraduate Excellence Scholarship, Mekelle Institute of Technology.

Professional Development

Deep Learning Specialization; Natural Language Processing Specialization; Machine Learning Engineering for Production Specialization (DeepLearning.AI / Coursera).